MIL-P-37407B
8 November 1985
SUPERSEDING
MIL-P-37407A
5 January 1977

MILITARY SPECIFICATION

PLUG, EAR, HEARING PROTECTION, CYLINDRICAL, UNIVERSAL SIZE, VINYL FOAM

This specification is approved for use within the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers disposable, polyvinyl chloride foam, cylindrical ear plugs suitable for use in hearing protection.

1.2 Classification.

Type I - Yellow
Type II - White

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the exist specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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AMSC N/A

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SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping Fiberboard.

MILITARY

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of specifications, standards and publication required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI S12.6-1984 Measurement of the Real-Ear Attenuation of Hearing Protectors at Threshold, Method for the.

(Application for copies should be addressed to the American Institute of Physics, 335 East 45th St., New York, NY 10017.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D568 Rate of Burning and/or Extent and Time of Burning of Flexible Plastics in a Vertical Position.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

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3.4 Sound attenuation. Sound attenuation shall be determined in accordance with the method described in 4.4.6. The mean attenuation shall exceed the following minimum values:

Frequency (Hz)	Mean Attenuation (dB)
125	26
250	29
500	30
1000	29
2000	33
3150	37
4000	37
6300	38
8000	37

3.5 Resistance to heat and humidity. The ear plug shall not deteriorate as shown by discoloration when aged as specified in 4.4.7.

3.6 Flammability. The ear plug shall not continue to burn for more than 10 seconds after removal of the flame when tested in accordance with 4.4.8. Furthermore, the ear plug shall not drip flaming particles while burning.

3.7 Patch test. There shall be no more than one positive reaction in the skin patch test conducted in accordance with 4.4.9.

3.8 Workmanship. The finished ear plug shall conform to the highest standards of commercial fabrication. The ear plug shall be free from defects which detract from its appearance or may impair its serviceability. The ear plug shall not cause irritation and shall be free from objectionable odor.

3.9 Instructions. Complete instructions with illustrations for the use and handling of the ear plugs shall be contained on the container for each pair.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The ear plug shall be fabricated of virgin polyvinyl chloride plastic foam which is non-dermatitic, non-toxic and non-allergenic.

3.2 Design and construction. The ear plug shall be smooth, of cylindrical shape, and have a diameter of 0.53 to 0.56 inch by a length of 0.71 to 0.85 inch when measured in accordance with the method of 4.4.1.

3.2.1 Density. The ear plug shall have a density of 0.080 to 0.108 gm/cc (5.0 to 6.75 lbs/ft^3) when examined in accordance with 4.4.2. However, no ear plug from any lot will vary more than $\pm$ 0.008 gm/cc ($\pm$ 0.5 lbs/ft^3) from the average for that lot.

3.2.2 Cell size. The ear plug shall have a small cell structure with no cell dimension exceeding 1.27 mm (0.050 inch) when examined in accordance with 4.4.3.

3.2.3 Color. Type I ear plugs shall be colored yellow and Type II ear plugs shall be white when examined in accordance with 4.4.4.

3.3 Recovery from compression. When tested as described in 4.4.5, the ear plug shall recover 90% of its original diameter in 5 to 60 seconds.

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4.1.1 Inspection. Inspection, as used in this specification is defined as both examination (such as visual or auditory investigation without the use of special laboratory applicances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificate of quality. Certificate of quality supplied by the manufacturer of the ear plug may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.3.1 Certification for repeat insult patch test. Certification in lieu of testing for the skin patch test is acceptable providing the testing has been done within the past three years in fulfillment of DPSC contract requirements, the record of usage is clean and the contractor ceritifies that the same raw materials and processing have been used for what is currently being offered.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purpose shall be one ear plug.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major	S-3	1.0
Minor	S-3	1.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II below. The unit of product for test purpose shall be one ear plug.

TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Dimensions	3.2	4.4.1	S-2	1.0
Density	3.2.1	4.4.2	S-2	1.0
Cell size	3.2.2	4.4.3	S-2	1.5
Color	3.2.3	4.4.4	S-2	2.5
Recovery from compression	3.3	4.4.5	S-3	1.0
Sound attenuation	3.4	4.4.6	S-3	1.0
Heat and Humidity	3.5	4.4.7	S-2	1.0
Flammability	3.6	4.4.8	S-1	1.0
Patch test	3.7	4.4.9	See 4.4.9	*

*Accept with no more than one positive reaction.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The ear plug shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and II. Examination, however, shall not be restricted to defects listed in table III. Any ear plug in the sample containing one or more defects shall be rejected and, if the number of defective ear plugs in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Major	
101	Not free of foreign material.
102	Edges not smooth, or presence of projections at edges.
103	Top and bottom surfaces not flat and parallel.
104	Top and bottom surfaces not perpendicular to long axis.
Minor	
201	Variations of color or shade within same plug.

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4.3.1 Inspection of packaging. The sampling and inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Dimensions. Dimensions of the ear plug shall be measured optically by means of an ocular micrometer-equipped microscope, tool maker's microscope or optical comporator.

4.4.2 Density. Density shall be determined by weighing the ear plug to 0.1 milligram and measuring the diameter and height to 0.025 millimeter. Density shall be expressed in grams/cubic centimeter or lbs/ft^3.

4.4.3 Cell size. Ear plugs shall be visually examined to find the largest cell size. Cell size shall be determined by means of a tool maker's microscope or a low power microscope with an accurately calibrated ocular micrometer.

4.4.4 Color. The Type I ear plug shall be yellow within the limits of Federal Standard 595a, Colors No. 33637 and 33793. The Type II ear plug shall be white and not darker than Federal Standard 595a Color No. 23619.

4.4.5 Recovery from compression. One-half of the ear plugs in the inspection lot specified in Table II shall be conditioned at 50% ± 5% relative humidity and the other half at 100% + 0%, -5%. The temperature shall be 70° to 75°F (21° to 24°C) and the time of exposure 24 hours. The ear plug is rolled and compressed between the thumb and forefinger for one minute while compressing it to its smallest diameter within the first 30 seconds and maintaining that diameter during the last 30 seconds. The ear plug shall then be dropped without delay into a glass tube with an inside diameter that is 90% (nominally 0.492 inch) of the ear plug's uncompressed diameter. The tube is held vertically and gently tapped on a hard surface until the ear plug expands to a diameter so that it is held by the tube. The recovery time is the time from the release of the ear plug from the fingers until it is held by the tube.

4.4.6 Sound attenuation. Sound attenuation shall be determined by the method of ANSI S12.6, Measurement of the Real-Ear Attenuation of Hearing Protectors at Threshold.

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4.4.7 Resistance to heat and humidity. The ear plug shall be exposed at 70° ± 2° C (158° ± 4° F) and 97% ± 3% relative humidity for 336 hours ± 2 hours.

4.4.8 Flammability. The ear plug shall be tested in accordance with ASTM Standard D568-77 except that the plug shall be suspended by means of a thin wire attached to the plug and the flame shall be applied for five seconds.

4.4.9 Repeat insult patch tests. Nine induction applications of a 25mm disc of the foam material used in the ear plugs shall be applied with tape on 100 human volunteers during the first three weeks of the test. The nine applications are made on alternate days, Monday, Wednesday and Friday, for 24-hour exposures. The patch areas are examined and scored 24 hours after removal of the patch of each of the nine applications. Following the last applications, the volunteers are "rested" for two weeks before the final 24-hour challenge application. The challenge patch exposure is examined and scored 24 and 72 hours after removal of the patch. A report shall be prepared stating the number of positive and negative reactions. Erythema of a distinct nature is a positive reaction.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Individual container. One pair of plugs shall be packaged in a suitable paperboard container of an appropriate size and design.

5.1.1.2 Unit package. Two hundred individually packaged pairs of plugs of one type only shall be packaged in a self-dispensing paperboard box having an adherent backing for attachment to a wall or bulkhead. Closure of the box shall be adequate to prevent spilling of contents under normal handling.

5.1.2 Level C. The level C preservation for ear plugs shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A. Ten unit packages containing plugs of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Ten unit packages containing plugs of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636 class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for ear plugs shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.3 Marking

5.3.1 Levels A, B and C.

5.3.1.1 Individual container. Each individual package shall be marked with instructions as specified in 3.9.

5.3.1.2 Unit package, exterior container and unitized load. Each unit package, exterior container and unitized load shall be marked as specified in MIL-STD-129. In addition, the lot or control number shall also be shown.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.3.1.

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5.2.3 Level C. The packaged plus shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

6. NOTES

6.1 Intended use. The ear plugs are intended for attenuation of high intensity sound and thereby protection of the ears.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) National Stock Number.

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-137-6345	PLUG, EAR, Hearing Protection, Cylindrical, Universal Size, Yellow, 400s
6515-01-117-7159	PLUG, EAR, Hearing Protection, Cylindrical, Universal Size, White, 400s

This specification does not include all types or sizes as indicated by the title nor all types which are commercially available.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

Project No. 6515-3582

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-37407B	PLUG, EAR, HEARING PROTECTION, CYLINDRICAL, UNIVERSAL SIZE, VINYL FOAM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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